

PUBLICATION LIST (2021-2026)

1. Interaction of Bile Salts, Surfactants, and Their Mixtures with the Phosphatidylcholine(d62) Lipid Monolayer/Water Interface Probed by VSFG Spectroscopy
Anisha Bandyopadhyay, Manidipa Basu and Jahur Alam Mondal
Langmuir, **2026**, 42, 8673-8681 (<https://pubs.acs.org/doi/10.1021/acs.langmuir.5c06737>)
2. Distinct Interfacial Behavior of Bile Salts and Its Interaction with Cetylpyridinium Chloride Cosurfactant at the Air/Water Interface
Anisha Bandyopadhyay, Manidipa Basu and Jahur Alam Mondal
Langmuir, **2025**, 41, 15190- 15198 (<https://pubs.acs.org/doi/10.1021/acs.langmuir.5c01922>)
3. Weak hydrogen bonds ($\pi\cdots\text{HOH}$, $\text{CH}\cdots\text{OH}_2$) in aqueous medium and their dependence on solute's charge polarity
Anisha Bandyopadhyay and Jahur Alam Mondal
J. Mol. Liq. **2025**, 127629 (<https://doi.org/10.1016/j.molliq.2025.127629>)
4. Vibrational Coupling and Hydrogen-Bond Structure of Water in the Extended Hydration Shell of Metal ions
Anisha Bandyopadhyay, Gunomoni Saha and Jahur Alam Mondal
J. Phys. Chem. B.,**2023**,127,7174-7180 (<https://pubs.acs.org/doi/10.1021/acs.jpcc.3c01883>)
5. Hydrogen-bonding and Vibrational Coupling of Water in a Hydrophobic Hydration Shell: Significance of Alkyl Chain Configuration and Charge on the Hydrophobe
Anisha Bandyopadhyay and Jahur Alam Mondal
J. Mol. Liq. **2023**,122987 (<https://doi.org/10.1016/j.molliq.2023.122987>)
6. Origin of Strong Hydrogen Bonding and Preferred Orientation of Water at Uncharged Polyethylene Glycol Polymer/Water Interface
Animesh Patra, **Anisha Bandyopadhyay**, Subhadip Roy and Jahur Alam Mondal
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7. Impact of electrolyte on the structure and orientation of water at air/water–polyethylene glycol polymer interface
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J. Chem. Phys., **2024**, 161, 174708 (<https://doi.org/10.1063/5.0231332>)
8. Intrinsic Raman scattering activity of OH and OD oscillators in liquid H₂O, D₂O, and HOD: Roles of zero-point energy and vibrational coupling
Sayanee Das, **Anisha Bandyopadhyay** and Jahur Alam Mondal
J. Chem. Phys., **2026**, 164, 244502 (<https://doi.org/10.1063/5.0330341>)
9. Unified view of the hydrogen-bond structure of water in the hydration shell of metal ions (Li^+ , Mg^{2+} , La^{3+} , Dy^{3+}) as observed in the entire 100–3800 cm^{-1} regions
Nishith Ghosh, **Anisha Bandyopadhyay**, Subhadip Roy, Gunomoni Saha and Jahur Alam Mondal
J. Mol. Liq., 389, **2023**, 122927 (<https://doi.org/10.1016/j.molliq.2023.122927>)
10. Vibrational Raman Spectroscopy of the Hydration Shell of Ions
Nishith Ghosh , Subhadip Roy, **Anisha Bandyopadhyay** and Jahur Alam Mondal
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11. Modulating the effective ionic radii of trivalent dopants in ceria using a combination of dopants to improve catalytic efficiency for the oxygen evolution reaction
Debarati Das, Jyoti Prakash, **Anisha Bandyopadhyay**, Annu Balhara, U. K. Goutam, Raghunath Acharya, Santosh K. Gupta and Kathi Sudarshan
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12. Electrophoretic deposition of CuO particulate thick film for ethanol sensing
Anisha Bandyopadhyay, Kamalika Mandal, Vibhav Ambardekar, Debasish Das and S. B. Majumder
J Mater Sci: Mater Electron., **2021**, 32,17324–17335 (<https://doi.org/10.1007/s10854-021-06247-0>)